Name	: Mrs. DEMO PATIENT	TID/SID	: UMR000000 / 10101010
Age/Gender	: 30 Years / Female	Registered On	: 01-Jan-2025 09:03 PM
Ref By	: Dr.Ravi Chenikala	Collected On	: 01-Jan-2025 09:10 PM
Reg.No	: DEMO123456	Reported On	: 02-Jan-2025 12:37 PM
		Reference	: Demo Reference

DEPARTMENT OF MOLECULAR PATHOLOGY
Non Invasive Prenatal Screening

I. Screening results

Chromosomes	Risk	Z score	Test Results	Reference interval
Chromosome 21 		-1.42	Low Risk	-6<Z score<2.8
Chromosome 18 		0.49	Low Risk	-6<Z score<2.8
Chromosome 13 		-2.90	Low Risk	-6<Z score<2.8
Sex Chromosomes		Part III	Low Risk	Part III
Other Chromosomes		Part IV	Low Risk	Part IV
Fetal fraction				
8.20%				

II. Supplementary information

1. The NIPS test screens a maternal blood sample for chromosome aneuploidy in placental DNA using the following methodology:
   (1) Extraction of cell-free placental DNA from the maternal blood sample
   (2) High throughput sequencing of the extracted cell-free placental DNA
   (3) Calculation of molecular mass of placental DNA in all chromosomes
2. The method is intended for use in pregnant women who are at least 10 weeks of pregnancy. The method is suitable for both singleton and twin pregnancies. The accuracy may be slightly lower in twin pregnancies due to multiple sources of fetal DNA.
3. Based on the scope, the NIPS test can detect the following:
   Whole Genome - 23 pairs of human chromosomes
   Sex chromosomal aneuploidies: XO, XXX, XXY/XYY
4. The test is capable of genome-wide aneuploidy detection over the whole fetal genome and gives the results for 23 pairs of chromosomes. This test confers an accuracy of up to 99% on the detection of fetal aneuploidy for chromosomes 13, 18 and 21. In a study of over 2000 samples, 6 samples were determined to be at high-risk of having an autosomal aneuploidy other than 13, 18 and 21. This is a prevalence rate of 0.3%, which is consistent with prevalence in published studies.

The sex of the fetus is not revealed as per PCPNDT act.

Results are indicated for screening, NOT diagnosis. – (Results should be reviewed and discussed with your healthcare provider.)

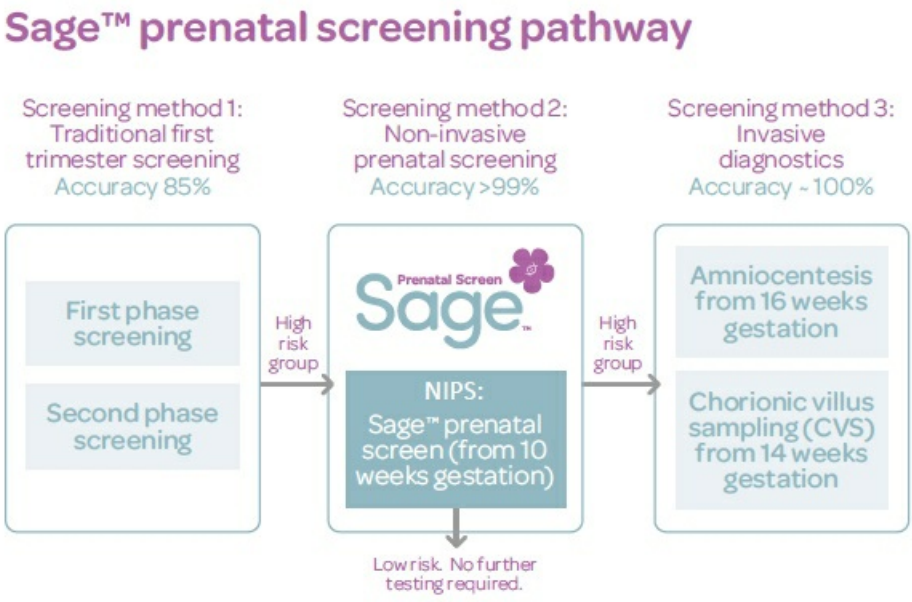
III. Sex Chromosomes

Sex Chromosome Aneuploidies	Risk	Z score	Test Results	Reference interval
XO			Low Risk	
XXY/XYY		0.54	Low Risk	Male -3<Z score<3 Female -2.8<Z score<2.8
XXX			Low Risk	

IV. Other Chromosomes

Chromosome	Risk	Z score	Test Results	Reference interval
Chromosome 1		-0.01	Low Risk	-6<Z score<6
Chromosome 2		0.60	Low Risk	-6<Z score<6
Chromosome 3		-0.14	Low Risk	-6<Z score<6
Chromosome 4		1.59	Low Risk	-6<Z score<6
Chromosome 5		-0.34	Low Risk	-6<Z score<6
Chromosome 6		0.36	Low Risk	-6<Z score<6
Chromosome 7		1.32	Low Risk	-6<Z score<6
Chromosome 8		1.24	Low Risk	-6<Z score<6
Chromosome 9		-1.38	Low Risk	-6<Z score<6
Chromosome 10		-0.65	Low Risk	-6<Z score<6
Chromosome 11		-1.31	Low Risk	-6<Z score<6
Chromosome 12		0.20	Low Risk	-6<Z score<6
Chromosome 14		0.89	Low Risk	-6<Z score<6
Chromosome 15		-0.64	Low Risk	-6<Z score<6
Chromosome 16		-0.93	Low Risk	-6<Z score<6
Chromosome 17		0.92	Low Risk	-6<Z score<6
Chromosome 19		0.92	Low Risk	-6<Z score<6
Chromosome 20		-1.46	Low Risk	-6<Z score<6
Chromosome 22		-1.12	Low Risk	-6<Z score<6

Pipeline version: 97bef21



About Sage™ prenatal screen

The Sage™ prenatal screen is a new advanced non-invasive prenatal screening solution using the latest developments in DNA technology to detect placental DNA in maternal blood. Sage™ offers a menu-based chromosome analysis to estimate the risk of afetus having Down's syndrome and other genetic disorders. Enabling pregnant women and their families fast, safe and reliable results and reducing the need for invasive tests and the associated risks, stress and anxiety. Sage™ is indicated for use in pregnant women who are at least 10-week pregnant. Chromosomal aneuploidy can then be detected using bioinformatics analyses, where the detection rate and sensitivity are over 99%.

Limitations

Sage™ is a screening test and all high-risk results should be confirmed through further investigation which may include tests such as amniocentesis or Chorionic Villus Sampling (CVS). Pregnant women with a high-risk result should be referred for genetic counseling and offered invasive prenatal diagnosis for confirmation of test results. Pregnant women with a negative test result do not ensure an unaffected pregnancy. While results of this testing are highly accurate, not all chromosomal abnormalities may be detected due to placental, maternal or fetal mosaicism, or other causes (micro-deletions, chromosome re-arrangements, translocations, inversions, unbalanced translocations, uniparental disomy). The test is not reportable for known multiple gestations, or if the gestational age is less than 10 weeks.

Test method

A simple maternal blood sample is taken from the pregnant mother from 10-week gestation with no risk to the fetus. Circulating cell-free placental DNA was purified from the plasma component of anti-coagulated 10mL of maternal whole blood. It was then converted into a genomic DNA library for Next Generation Sequencing and then determination of chromosomal aneuploidy.

References:

1. Obstet Gynecol 2012;119:890-901.
2. BMJ 2011;342:c7401.
3. Prenat Diagn 2012;32:c7401.
4. ACOG/SMFM Joint Committee Opinion No. 545, Dec 2012.

Manager Molecular Diagnostics	Consultant Geneticist	HOD Molecular Diagnostics